§11.2 Series

Homework：10,17,33,39,43,45,46,49,57,62,63,65

Definition： Series $=\sum_{i=1}^{\infty} a_i = a_1 + a_2 + \ldots\ldots + a_n + \ldots\ldots$

Convergence/Divergence of a Series.

(nth) partial sum $= S_n = \sum_{i=1}^{n} a_i$

$\{S_n\}$ converges / diverges $\Rightarrow \sum_{i=1}^{\infty} a_i$ converges / diverges.

Theorem1：Geometric Series.

$\sum_{n=1}^{\infty} r^n$ converges $\Leftrightarrow$ $|r|<1$ and $\sum_{n=1}^{\infty} r^n = \frac{r}{1-r}$.

Theorem2： $\sum_{n=1}^{\infty} a_n$ converges $\Rightarrow \lim_{n\to\infty} a_n = 0$.

($\lim_{n\to\infty} a_n \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ diverges.)

Remark： $\lim_{n\to\infty} \frac{1}{n} = 0$, but $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

$$\because \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3}+\frac{1}{4}\right)}_{>\frac{1}{2}} + \underbrace{\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)}_{>\frac{1}{2}} + \underbrace{\left(\frac{1}{9}+\ldots+\frac{1}{16}\right)}_{>\frac{1}{2}} + \ldots$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Example 1：

(i) $\sum_{n=1}^{\infty} (-1)^n \Rightarrow$ diverges.

(ii) $\sum_{n=1}^{\infty} \frac{\ln n}{n} \Rightarrow$ diverges. $\left(\because \frac{\ln n}{n} > \frac{1}{n} \text{ when } n \geq 3\right)$.

(iii) $\sum_{n=1}^{\infty} \frac{n}{n+1} \Rightarrow$ diverges $\left(\because \frac{n}{n+1} \to 1 \text{ as } n \to \infty\right)$.

Example 2： $\sum_{n=1}^{\infty} 2^{2n} 3^{1-n} = \sum 4^n \cdot 3^{(-n)} \cdot 3 = \sum 3 \cdot \left(\frac{4}{3}\right)^n$

$r = \frac{4}{3} > 1 \Rightarrow$ divergence.

Example 3：

The Cantor Let , named after the German mathematician Georg Cantor (1845-1918), is constructed as follows. We start with the closed interval $[0,1]$ and remove the open interval $\left(\frac{1}{3}, \frac{2}{3}\right)$. That leaves the two intervals $\left(0, \frac{1}{3}\right)$ and $\left(\frac{2}{3}, 1\right)$, and we remove the open middle third of each. We continue this procedure indefinitely, at each step removing the open middle third of every interval that remains from the proceeding step. The Cantor set consists of the number that remain in $[0,1]$ after all those intervals have been removed.

(a) Show that the total length of all the intervals that are removed is 1.

(b) Give examples of some numbers in the Cantor set.

Proof：

(a)

$$
\begin{aligned}
S &= \frac{1}{3} + 2\left(\frac{1}{3}\right)^2 + 2^2\left(\frac{1}{3}\right)^3 + \ldots + 2^n\left(\frac{1}{3}\right)^{n+1} + \ldots\ldots \\
&= \frac{1}{3}\left(1 + \left(\frac{2}{3}\right) + \left(\frac{2}{3}\right)^2 + \ldots + \left(\frac{2}{3}\right)^n + \ldots\ldots\right) \\
&= \frac{1}{3}\frac{1}{1-\frac{2}{3}} \\
&= 1
\end{aligned}
$$

(b) $\frac{1}{3}, \frac{2}{3}, \frac{1}{9}, \frac{2}{9}, \frac{7}{9}, \frac{8}{9}, \ldots\ldots$

Example 4：

The Sierpinski Carpet is a 2-dimensional counterpart of the Cantor set. We begin with a square of side 1. We remove the center one-ninth of a square. The process is then repeated just like the process done in Example3.